Xin Ma



Research interests: theoretical foundations and interpretability of machine learning and large language models.

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Education

University of Science and Technology of China 2024.09 – Present
M.S. in Data Science (Statistics), School of Artificial Intelligence and Data Science

South China University of Technology 2020.09 – 2024.06
B.S. in Mathematics and Applied Mathematics (Statistics), School of Mathematics

Selected Awards

- **Graduate Academic Scholarship**, USTC (First Class 2025, Second Class 2026)
- **Outstanding Graduate**, South China University of Technology (2024)
- **National Encouragement Scholarship** (2021–2024)
- **National Scholarship**, Ministry of Education of China (2020)

Publications

1. (2026) "More Edits, More Stable: Understanding the Lifelong Normalization in Sequential Model Editing."
Xin Ma, Wei Chen, Qi Liu, Derong Xu, Zhi Zheng, Tong Xu, Enhong Chen.
The 43rd International Conference on Machine Learning (ICML 2026).

Research Experience

Understanding Lifelong Normalization and Designing a Stable Sequential Model Editor (StableEdit) 2025.06 – 2026.01
First author | Accepted to ICML 2026

- **Problem.** Sequential model editing injects new facts over long horizons but degrades through catastrophic forgetting and model collapse. Recent editors that stay stable at the million-edit scale (e.g., UltraEdit, RLEdit) all share one unexplained component, **Lifelong Normalization (LN)**, whose removal collapses performance.
- **Theory (core contribution).** Gave the first theoretical account of LN: its online statistics amount to recursive Bayesian estimation of a drifting gradient distribution under a Normal–Inverse–Wishart prior, and the resulting updates provably have **bounded norm** and **asymptotic orthogonality**—jointly explaining the counterintuitive "more edits, more stable" effect.
- **Method & results.** Turned this analysis into **StableEdit** (statistics warm-up and full-covariance whitening). Across four LLMs and four benchmarks it stays stable up to **2M edits**, improving generalization by ∼**7.4 points** over UltraEdit at only **1.43**× the fastest baseline's runtime.

Theoretical Analysis of a Selective Supervised Fine-Tuning Method for LLMs 2025.12 – 2026.01
Collaborative project — responsible for the theoretical mechanism and core intuition

- **Setting & method.** In the SFT-then-RL pipeline, uniform cross-entropy fine-tuning narrows the output distribution and triggers entropy collapse, weakening later RL exploration. The project instead masks high-entropy and high-KL-divergence tokens and regularizes them with entropy and KL terms, preserving diversity.
- **Contribution (theory).** Explained the mechanism through a token-level gradient analysis: cross-entropy keeps a one-hot forcing term on uncertain tokens that drives entropy collapse and parameter drift, whereas the masked, label-free update suppresses it; also showed why selective regularization outperforms global regularization.
- **Evidence.** On Qwen3-8B the method shifts fewer parameters from the base model than standard SFT (≈**22.6%** vs. ≈**27.2%**), with ablations confirming both regularizers are essential and higher pass@k indicating stronger exploration.

Skills

Programming: Python, C++, MATLAB, R
Writing & typesetting: LaTeX, Markdown
Scientific plotting: Matplotlib and various scientific visualization tools